Advanced Number Theory

Divisibility rules and prime factorization

1 Find the prime factorization of 1260.

2 Determine whether 143 is prime, and justify your answer.

GCF, LCM, and their relationship

3 Find the GCF and LCM of 36 and 60.

4 Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity GCF × LCM = product of the two numbers.

Modular arithmetic

5 Find the remainder when 47 is divided by 6, and express this using modular notation.

6 Find the last digit of 7^{100}, using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).

Sums and products of integer sets

7 Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.

8 Find the sum of all even integers from 2 to 200.

Number sets and properties

9 Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.

10 Explain why the sum of a rational number and an irrational number is always irrational.

Sequences and series

11 Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.

12 Find the sum of the first 8 terms of the geometric sequence $2, 6, 18, 54, \ldots$

Absolute value and inequalities with integers

13 Find all integer solutions of $|x - 5| \leq 3$.

Combining number theory concepts in a multi-part problem

14 This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity GCF $\times$ LCM = product of the two numbers.

Perfect squares, perfect cubes, and their properties

15 Find the smallest positive integer n such that $200n$ is a perfect square.

16 Determine whether 128 is a perfect cube, and justify your answer using prime factorization.

Counting divisors of a number

17 Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.

Working with consecutive integers and integer constraints

18 Find three consecutive even integers whose sum is 96.

19 The product of two consecutive positive integers is 156. Find the integers.

Digit sum and place value problems

20 A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.

A multi-part modular arithmetic and remainder problem

21 This question has three parts. (a) Find the remainder when 2^{10} is divided by 7. (b) Use the pattern of remainders of powers of 2 divided by 7 (which cycle) to find the remainder when 2^{50} is divided by 7. (c) Explain the general method for finding remainders of large powers using cycles.